Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/17/2017
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Project Title:	NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	01/01/2016	**End Date:**	12/31/2020
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	NOTE: Continuation of "NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL," grant number NNX10AU19G. Funding is for Dr. Kelton's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main project: Thermophysical and structural properties and solidification behavior of supercooled glass forming liquids as a member of THERMOLAB; the Science Coordinator is Prof. Dr. Hans Fecht, Ulm University, Institute of Micro and Nanomaterials, Albert-Einstein-Allee 47, Ulm Germany. Similar measurements on undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); the Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany. Subprojects: 1. Ground-Based Experimental Program: Synchrotron X-ray studies of liquid structures are made at the Advanced Photon Source using the ESL (electrostatic levitation) chamber at Washington University (WU-BESL), which has been optimized for scattering studies. These are correlated with thermophysical property data obtained on ground with WU-BESL and using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) on the International Space Station (ISS). Terrestrial maximum undercooling and induction time to recalescence data are obtained in ESL, where stirring ensures that nucleation is not diffusion controlled. This data will allow an evaluation of diffusion in the upcoming MSL-EML ISS undercooling studies. 2. Thermophysical Property Measurements: ISS. Specific heat, density, surface tension and viscosity measurements will be made of supercooled glass-forming liquids and of Ti-Zr-Ni liquids with different Ni concentrations as a function of the temperature using the MSL-EML facility. These will be correlated with x-ray synchrotron diffraction studies of the supercooled liquids and ground-based measurements made using WU-BESL The purpose is to determine the existence and nature of liquid/liquid phase transitions that may be associated with chemical and topological ordering and a possible connection with liquid fragility as well as to obtain indicators in the specific heat of a cooperative cross over behavior in the shear viscosity at high temperatures. 3. Effect of diffusion on nucleation: The importance of long-range diffusion on nucleation will be evaluated using the MSL-EML facility located on the ISS. The critical items of interest are the maximum undercooling and the induction time to recalescence in isothermal holds at temperatures below the liquidus temperature. Together with the results from the thermophysical property measurements and an MHD convection model results this will allow us to analyze how the maximum undercooling is influenced by fluid flow. These results of the micro-g experiments will be compared with the results of ground-based ESL measurements and with predictions from Kelton’s coupled-flux model for nucleation. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: 1. Ground Based Experiment Program. Measurements of the density, surface tension and viscosity and synchrotron x-ray diffraction measurements of the liquid structure in the supercooled state using WU-BESL. Ground-based measurements of the maximum undercooling and induction time to recalescence in isothermal holds will also be made using WU-BESL. These will be used to evaluate the nature of ordering on phase transitions in the supercooled liquid and fragility. 2. Long Duration Microgravity Experiments. Measurements of thermophysical properties, specific heat (modulated measurements), surface tension, viscosity and density, and also electrical resistivity will be made as a function of temperature using the MSL-EML on the ISS. These will be coupled with x-ray and elastic neutron diffraction measurements of the liquid structure factor (WP 2) and ground based studies of the density and viscosity to study the role of ordering on novel liquid/liquid phase transitions and to determine the nature of these transitions. Maximum undercooling and time to recalesce measurements as a function of temperature and stirring will be compared with terrestrial ESL measurements to assess the importance of diffusion on nucleation, allowing Kelton's coupled flux model for nucleation to be evaluated experimentally.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which have the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science, encapsulated in the Materials Genome Initiative. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. Even containerless based methods, such as electrostatic levitation, cannot be used to accurately investigate these. The ISS experiments that are being carried out under the support of this grant are allowing these data to be obtained.

Task Progress:	Following is a brief summary of the activities carried out under NASA grant NNX16B52G over the past year. Experiments on the International Space Station (ISS) -- ISS experiments using the EML-MSL facility were conducted during the period May 30 to July 17, 2017 on three batch 2 samples: Ti39.5Zr39.5Ni21 (ID#39), Zr64Ni36 (ID#36), and Cu50Zr50 (ID#27). Representatives of the collaborative teams of QUASI, ICOPROSOL, ELFSTONE, and MULTIPHASE took part in those experiments. The experiments included maximum undercooling and growth measurements as a function of stirring, and measurements of the specific heat and electrical resistivity in the equilibrium and supercooled liquids. The Ti39.5Zr39.5Ni21 liquid showed a greater undercooling in ground-based electrostatic levitation (ESL) studies than in the electromagnetic levitation (EML) studies on the ISS. This could be consistent with the predictions of the coupled-flux model developed in Kelton's group, which was a key goal of the ISS experiments. However, the larger EML sample cools more slowly than the ESL ones, which could also contribute to the lesser undercooling. Stirring calculations by the USTIP are underway to check this. Crystal growth velocity measurements were made on Ti39.5Zr39.5Ni21 and Zr64Ni36. The growth rate of the structurally complex icosahedral phase from the liquid of Ti39.5Zr39.5Ni21 is extremely slow. Little is known about the growth rates of such complex structures; these measurements are, therefore, important for developing that understanding. The electrical resisitivities of these liquids were measured for the first time. For all three samples, both the measured resistivity and the specific heat show abrupt increases with decreasing temperature close to a temperature, TA, where a crossover behavior is observed in the viscosity. Molecular dynamics calculations explain the crossover as arising from the onset of cooperativity in the liquid and have shown that TA is intimately related to the glass transition temperature, Tg, which we have confirmed experimentally. The results obtained on the ISS are very exciting, connecting mass transport with charge transport for the first time and showing that the crossover in dynamics is fundamentally linked with many other physical properties of the liquid. Other activities with ESA linked to the ISS experiments -- The Principal Investigator (PI) participated in the International Working Group (IWG) 16 meeting, at Noordwijk, Netherlands, on 28-29 August, 2017, to present some of the results from obtained in his Batch 2 studies discussed above. Concern was also raised about continuing contamination issues in the gas environments of the MSL-EML. Ground-based activities in support of the ISS experiments -- Significant results were obtained in ground-based studies that support the ISS experiments. Some of the key results are listed below: 1. Undercooling measurements of Ti39.5Zr39.5Ni21 -- Over one hundred undercooling measurements were made in the ESL located in the PI's lab for a statistical analysis to obtain estimates of the nucleation barrier and the kinetic pre-term for comparison with the data obtained on the ISS. 2. Liquid fragility and the character of the interatomic potential -- While the fragility of a liquid, described by the temperature dependence of the viscosity, is easily measured and is important for glass formation and crystallization, the microscopic origin of fragility is poorly understood. In the past, we demonstrated that it is related to the cohesive energy and to the anharmonicity of the interatomic potential, obtained from thermal expansion measurements. In the past year, we finished experimental studies and published the results in a paper in Nature Materials that shows it is also related to the strength of the repulsive part of the potential. Taking all of these studies together, it is clear that the form of the interatomic potential determines fragility. One of the strongest liquids is SiO2 with strong covalent bonding. While the bonding is not covalent in metallic liquids, the form of the interatomic potential acts to mimic this. 3. Thermal expansion coefficient -- Our studies have shown that for liquids, the volume expansion obtained from direct measurements in the ESL and from shifts in the positions of the first peak in the structure factor, S(q), do not agree, although they do for crystalline materials. To resolve this using a dilatometer we measured the expansion coefficients in metallic glasses of interest for the ISS studies and compared the results with the coefficients obtained from S(q). They are in good agreement for the glasses, indicating that the disagreement with the liquids is due to their change in structure with temperature. This resolves a source of controversy that has existed in the metallic glass community for over a decade. 4. Pair correlation function and viscosity -- Using the two particle correlation function (g(r)), obtained from our X-ray scattering experiments, we have estimated the excess entropy and found that it compares with the measured viscosity. This is another demonstration of an intimate connection between structure and dynamics, which has been a recurrent theme in our ground-based research over several years.

Bibliography Type:	Description: (Last Updated: 05/20/2025)
Articles in Peer-reviewed Journals	Pueblo CE, Sun M, Kelton KF. "Strength of the repulsive part of the interatomic potential determines fragility in metallic liquids." Nature Materials. 2017 Aug;16(8):792-6. https://doi.org/10.1038/nmat4935 ; PubMed PMID: 28692041 , Aug-2017
Articles in Peer-reviewed Journals	Gangopadhyay AK, Kelton KF. "Recent progress in understanding high temperature dynamical properties and fragility in metallic liquids, and their connection with atomic structure." Journal of Materials Research. 2017 Jul 28;32(14):2638-57. https://doi.org/10.1557/jmr.2017.253 , Jul-2017
Articles in Peer-reviewed Journals	Gangopadhyay AK, Pueblo CE, Dai R, Johnson ML, Ashcraft R, Van Hoesen D, Sellers M, Kelton KF. "Correlation of the fragility of metallic liquids with the high temperature structure, volume, and cohesive energy." J Chem Phys. 2017 Apr 21;146(15):154506. https://doi.org/10.1063/1.4981011 ; PubMed PMID: 28433017 , Apr-2017
Articles in Peer-reviewed Journals	Lee GW, Cho YC, Lee, B, Kelton KF. "Interfacial free energy and medium range order: Proof of an inverse of Frank's hypothesis." Physical Review B. 2017 Feb 1;95:054202 (7 p.). https://doi.org/10.1103/PhysRevB.95.054202 , Feb-2017
Articles in Peer-reviewed Journals	Kelton KF. "Kinetic and structural fragility—a correlation between structures and dynamics in metallic liquids and glasses." J Phys Condens Matter. 2017 Jan 18;29(2):023002. Epub 2016 Nov 14. https://doi.org/10.1088/0953-8984/29/2/023002 ; PubMed PMID: 27841996 , Jan-2017
Articles in Peer-reviewed Journals	Gangopadhyay AK, Kelton KF. "Measurements of the temperature dependent total hemispherical emissivity using an electrostatic levitation facility." International Journal of Thermophysics. 2017 Jan;38(1):3 (8 p.). https://doi.org/10.1007/s10765-016-2138-x , Jan-2017